

CS2220: Introduction to Computational Biology

Sample Pairwise Alignment

Somayyeh Koohi

Fall 2024



Pairwise Alignment

Sequence A: CTTAACT

Sequence B: CGGATCAT

An alignment of A and B:

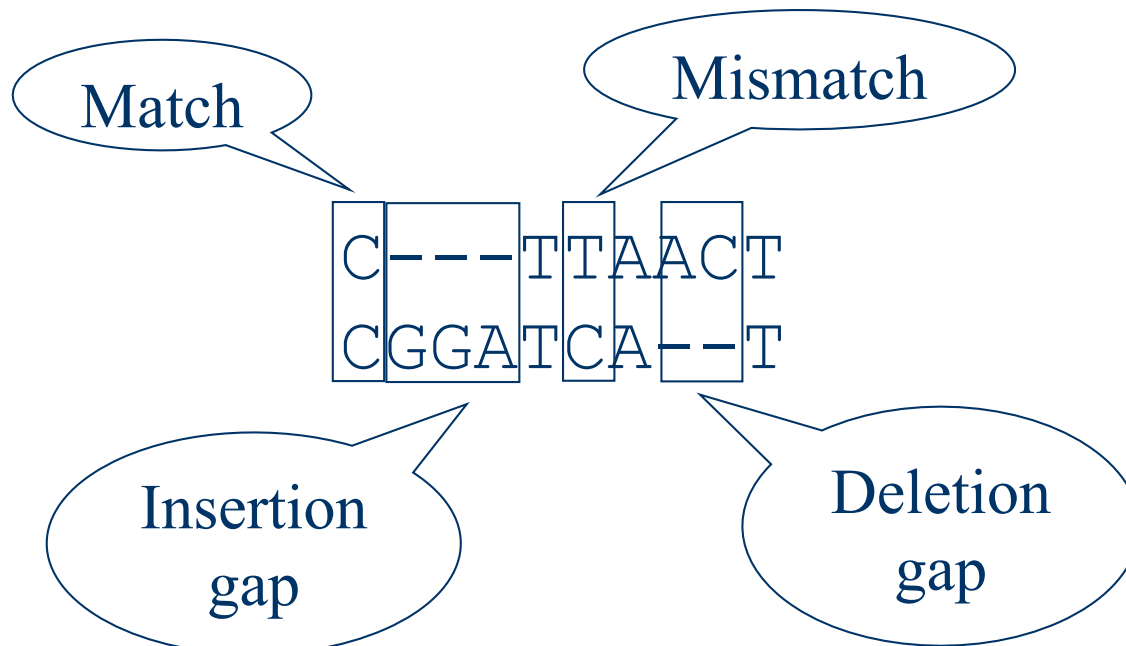
C---	TTAACT	←	Sequence A
CGGATCA	--T	←	Sequence B

Pairwise Alignment

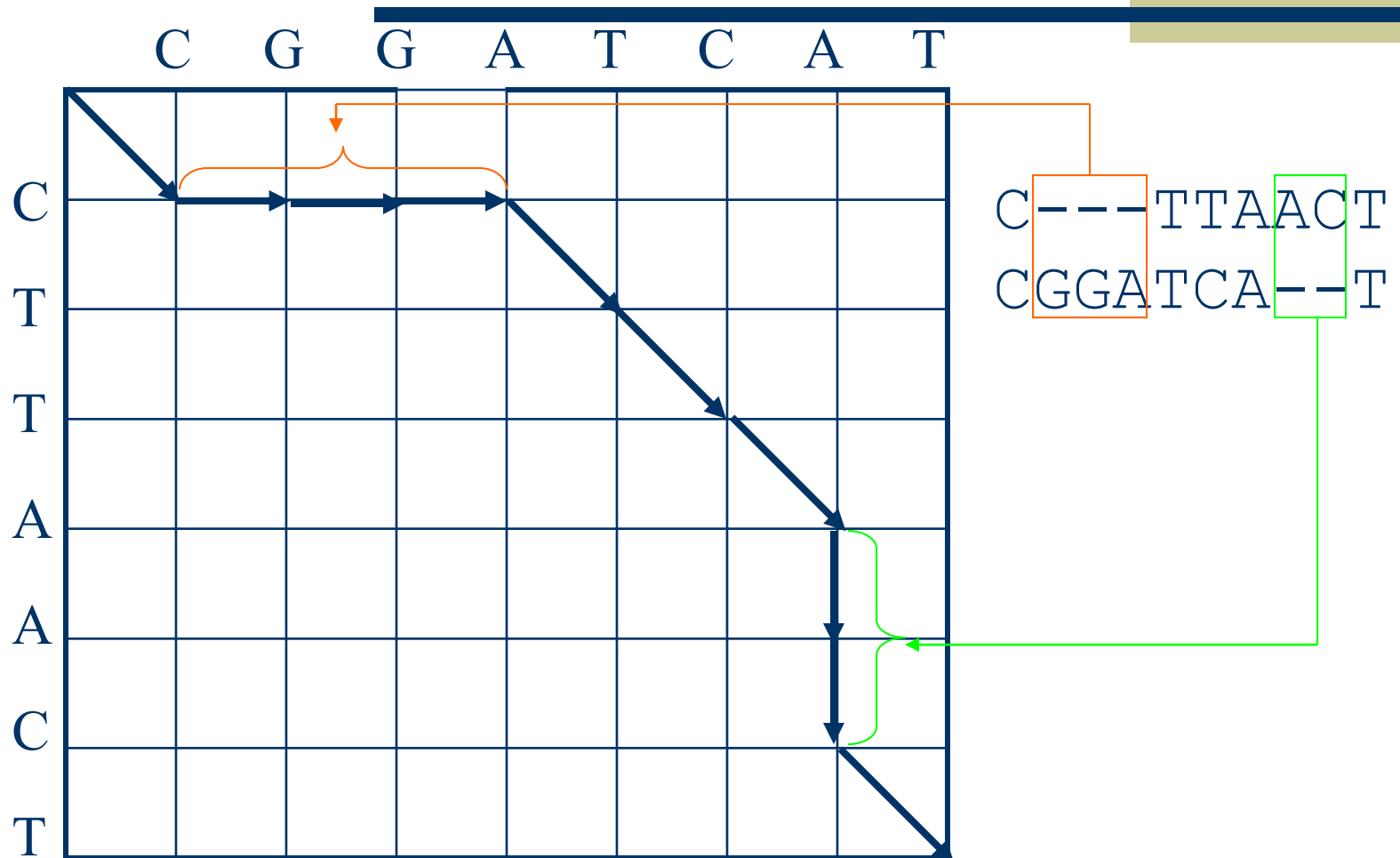
Sequence A: CTTAACT

Sequence B: CGGATCAT

An alignment of A and B:



Alignment Graph



A simple scoring scheme

- Match: +8 ($w(x, y) = 8$, if $x = y$)
- Mismatch: -5 ($w(x, y) = -5$, if $x \neq y$)
- Each gap symbol: -3 ($w(-, x) = w(x, -) = -3$)

C	-	-	-	T	T	A	A	C	T	
C	G	G	A	T	C	A	-	-	T	
+8	-3	-3	-3	+8	-5	+8	-3	-3	+8	= +12

Alignment score

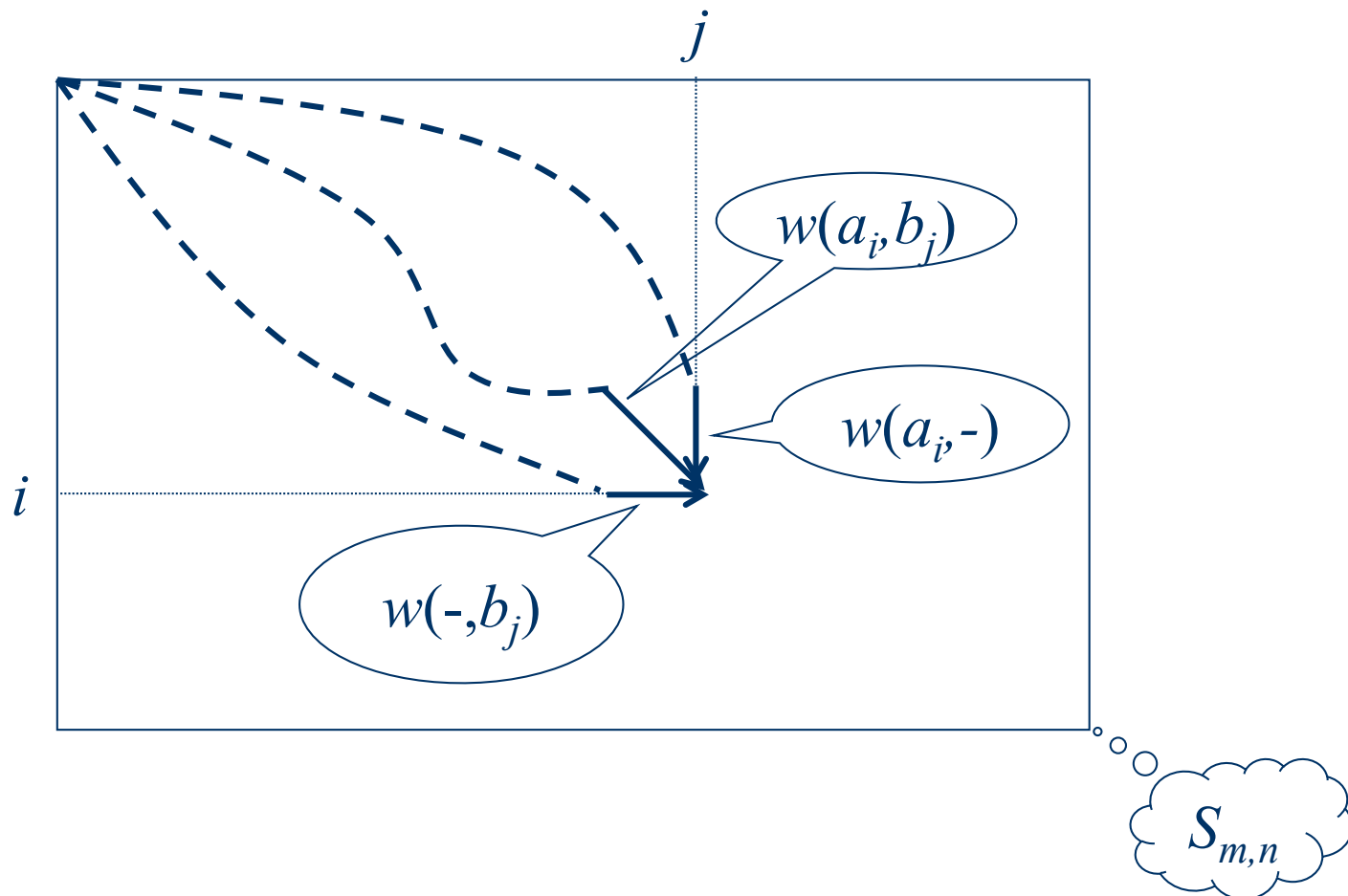
An optimal alignment

-- the alignment of maximum score

- Let $A=a_1a_2\dots a_m$ and $B=b_1b_2\dots b_n$.
- $S_{i,j}$: the score of an optimal alignment between $a_1a_2\dots a_i$ and $b_1b_2\dots b_j$
- With proper initializations, $S_{i,j}$ can be computed as follows.

$$s_{i,j} = \max \begin{cases} s_{i-1,j} + w(a_i, -) \\ s_{i,j-1} + w(-, b_j) \\ s_{i-1,j-1} + w(a_i, b_j) \end{cases}$$

Computing $S_{i,j}$



Initializations

		C	G	G	A	T	C	A	T
C T T A A C T	0	-3	-6	-9	-12	-15	-18	-21	-24
	-3								
	-6								
	-9								
	-12								
	-15								
	-18								
	-21								

$$S_{3,5} = ?$$

		C	G	G	A	T	C	A	T
C T T A A C T	0	-3	-6	-9	-12	-15	-18	-21	-24
	-3	8	5	2	-1	-4	-7	-10	-13
	-6	5	3	0	-3	7	4	1	-2
	-9	2	0	-2	-5	?			
	-12								
	-15								
	-18								
	-21								

$$S_{3,5} = 5$$

		C	G	G	A	T	C	A	T	
C T T A A C T		0	-3	-6	-9	-12	-15	-18	-21	-24
		-3	8	5	2	-1	-4	-7	-10	-13
		-6	5	3	0	-3	7	4	1	-2
		-9	2	0	-2	-5	5	-1	-1	9
		-12	-1	-3	-5	6	3	0	7	6
		-15	-4	-6	-8	3	1	-2	8	5
		-18	-7	-9	-11	0	-2	9	6	3
		-21	-10	-12	-14	-3	8	6	4	14

optimal
score

C T T A A C - T

C G G A T C A T

$$8 - 5 - 5 + 8 - 5 + 8 - 3 + 8 = 14$$

		C	G	G	A	T	C	A	T	
C T T A A C T	C	0	-3	-6	-9	-12	-15	-18	-21	-24
	T	-3	8	5	2	-1	-4	-7	-10	-13
	T	-6	5	3	0	-3	7	4	1	-2
	A	-9	2	0	-2	-5	5	-1	-1	9
	A	-12	-1	-3	-5	6	3	0	7	6
	C	-15	-4	-6	-8	3	1	-2	8	5
	C	-18	-7	-9	-11	0	-2	9	6	3
	T	-21	-10	-12	-14	-3	8	6	4	14

Now try this example in class

Sequence A: CAATTGA

Sequence B: GAATCTGC

Their optimal alignment ?

Initializations

		G	A	A	T	C	T	G	C
C A A T T G A	0	-3	-6	-9	-12	-15	-18	-21	-24
	-3								
	-6								
	-9								
	-12								
	-15								
	-18								
	-21								

$$S_{4,2} = ?$$

		G	A	A	T	C	T	G	C
C A A T T G A	0	-3	-6	-9	-12	-15	-18	-21	-24
	-3	-5	-8	-11	-14	-4	-7	-10	-13
	-6	-8	3	0	-3	-6	-9	-12	-15
	-9	-11	0	11	8	5	2	-1	-4
	-12	-14	?						
	-15								
	-18								
	-21								

$$S_{5,5} = ?$$

		G	A	A	T	C	T	G	C
C A A T T G A	0	-3	-6	-9	-12	-15	-18	-21	-24
	-3	-5	-8	-11	-14	-4	-7	-10	-13
	-6	-8	3	0	-3	-6	-9	-12	-15
	-9	-11	0	11	8	5	2	-1	-4
	-12	-14	-3	8	19	16	13	10	7
	-15	-11	-6	5	16	?			
	-18								
	-21								

$$S_{5,5} = 14$$

		G	A	A	T	C	T	G	C
C A A T T G A	0	-3	-6	-9	-12	-15	-18	-21	-24
	-3	-5	-8	-11	-14	-4	-7	-10	-13
	-6	-8	3	0	-3	-6	-9	-12	-15
	-9	-11	0	11	8	5	2	-1	-4
	-12	-14	-3	8	19	16	13	10	7
	-15	-11	-6	5	16	14	24	21	18
	-18	-7	-9	2	13	11	21	32	29
	-21	-10	1	-1	10	8	18	29	27

optimal
score

C A A T - T G A
 G A A T C T G C
 $-5 +8 +8 +8 -3 +8 +8 -5 = 27$

	G	A	A	T	C	T	G	C	
C	0	-3	-6	-9	-12	-15	-18	-21	-24
	-3	-5	-8	-11	-14	-4	-7	-10	-13
A	-6	-8	3	0	-3	-6	-9	-12	-15
A	-9	-11	0	11	8	5	2	-1	-4
T	-12	-14	-3	8	19	16	13	10	7
T	-15	-11	-6	5	16	14	24	21	18
G	-18	-7	-9	2	13	11	21	32	29
A	-21	-10	1	-1	10	8	18	29	27

An optimal local alignment

- $S_{i,j}$: the score of an optimal local alignment ending at a_i and b_j
- With proper initializations, $S_{i,j}$ can be computed as follows.

$$s_{i,j} = \max \begin{cases} 0 \\ s_{i-1,j} + w(a_i, -) \\ s_{i,j-1} + w(-, b_j) \\ s_{i-1,j-1} + w(a_i, b_j) \end{cases}$$

local alignment

		C	G	G	A	T	C	A	T
C	0	0	0	0	0	0	0	0	0
T	0	8	5	2	0	0	8	5	2
T	0	5	3	0	0	8	5	3	13
A	0	2	0	0	0	8	5	2	11
A	0	0	0	0	8	5	3	?	
C	0								
T	0								

Match: 8

Mismatch: -5

Gap symbol: -3

local alignment

		C	G	G	A	T	C	A	T
C	0	0	0	0	0	0	0	0	0
	0	8	5	2	0	0	8	5	2
	0	5	3	0	0	8	5	3	13
	0	2	0	0	0	8	5	2	11
A	0	0	0	0	8	5	3	13	10
A	0	0	0	0	8	5	2	11	8
C	0	8	5	2	5	3	13	10	7
T	0	5	3	0	2	13	10	8	18

Match: 8

Mismatch: -5

Gap symbol: -3

The
best
score_{*l*}

A - C - T

A T C A T

$$8 - 3 + 8 - 3 + 8 = 18$$

		C	G	G	A	T	C	A	T
C	0	0	0	0	0	0	0	0	0
T	0	8	5	2	0	0	8	5	2
T	0	5	3	0	0	8	5	3	13
A	0	2	0	0	0	8	5	2	11
A	0	0	0	0	8	5	3	13	10
C	0	0	0	0	8	5	2	11	8
C	0	8	5	2	5	3	13	10	7
T	0	5	3	0	2	13	10	8	18

The
best
score

Now try this example in class

Sequence A: CAATTGA

Sequence B: GAATCTGC

Their optimal local alignment ?

Did you get it right?

		G	A	A	T	C	T	G	C	
C A A T T G A	C	0	0	0	0	0	0	0	0	
	A	0	0	0	0	8	5	2	8	
	A	0	0	8	8	5	5	3	0	5
	T	0	0	8	16	13	10	7	4	2
	T	0	0	5	13	24	21	18	15	12
	G	0	0	2	10	21	19	29	26	23
	G	0	8	5	7	18	16	26	37	34
	A	0	5	16	13	15	13	23	34	32

A A T - T G

A A T C T G

$$8+8+8-3+8+8 = 37$$

		G	A	A	T	C	T	G	C	
C		0	0	0	0	0	0	0	0	
	C	0	0	0	0	8	5	2	8	
A	A	0	0	8	8	5	5	3	0	5
A	A	0	0	8	16	13	10	7	4	1
T	T	0	0	5	13	24	21	18	15	12
T	T	0	0	2	10	21	19	29	26	23
G	G	0	8	5	7	18	16	26	37	34
A	A	0	5	16	13	15	13	23	34	32

Useful Websites

- **MIT Biology Hypertextbook**
 - <http://www.mit.edu:8001/afs/athena/course/other/esgbio/www/7001/main.html>
- **The International Society for Computational Biology:**
 - <http://www.iscb.org/>
- **National Center for Biotechnology Information (NCBI, NIH):**
 - <http://www.ncbi.nlm.nih.gov/>
- **European Bioinformatics Institute (EBI):**
 - <http://www.ebi.ac.uk/>
- **DNA Data Bank of Japan (DDBJ):**
 - <http://www.ddbj.nig.ac.jp/>